

Unit 4: One-Sided Limits

1. The idea, in plain words

So far we've asked "what does $f(x)$ approach as x gets close to a ?" — sneaking up on a from both directions at once. But sometimes a function behaves differently depending on which side you sneak up from. One-sided limits let us check each direction separately.

Picture it: imagine walking along a graph toward a certain x -value, a .

- If you're walking in from the left (from smaller x -values, heading up toward a), the height you're approaching is called the left-hand limit, written $\lim_{x \rightarrow a^-} f(x)$.
- If you're walking in from the right (from larger x -values, heading down toward a), the height you're approaching is called the right-hand limit, written $\lim_{x \rightarrow a^+} f(x)$.

Most of the time — for nice smooth curves — it doesn't matter which side you approach from; you land at the same height either way. But piecewise functions (functions built out of different formulas glued together at a breakpoint) can behave very differently on the two sides. Imagine a staircase: walking up to a step from the left, you're at one height; stepping in from the right, you might be at a completely different height. In that case, there's no single "the limit" — the two sides disagree.

The big rule: the ordinary (two-sided) limit only exists if both one-sided limits exist and agree with each other. If they don't match, we say the limit "does not exist" (DNE) at that point.

Important reminder: a limit only cares about what $f(x)$ is doing near a — it does not care what $f(a)$ itself equals, or even whether f is defined at a at all. Don't let the actual value at the point distract you from what the function is approaching.

2. Toolbox

Left-hand limit: consider only x -values slightly less than a .

$$\lim_{x \rightarrow a^-} f(x)$$

Right-hand limit: consider only x -values slightly greater than a .

$$\lim_{x \rightarrow a^+} f(x)$$

Existence rule:

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L$$

If the two one-sided limits are different numbers, the two-sided limit does not exist.

For piecewise functions: use whichever formula applies just below a to compute the left-hand limit, and whichever formula applies just above a to compute the right-hand limit. Then plug a into each formula (direct substitution, or factor-and-cancel first if needed, same as Unit 2).

For absolute value expressions, remember:

$$|x - a| = \begin{cases} x - a, & x \geq a \\ -(x - a), & x < a \end{cases}$$

This is exactly why absolute value functions are a classic source of one-sided limit problems — the formula genuinely changes depending on which side you're on.

3. Common mistakes

- Grabbing the wrong piece. Always double check: which formula is valid for x -values just below a ? Which one for just above? It's easy to grab the wrong one under time pressure.
- Stopping after checking only one side. To claim the two-sided limit exists, you must check both sides and confirm they match — don't skip the second side.
- Confusing the limit with the function's actual value at a . A piecewise function often defines a specific (sometimes weird) value right at the breakpoint — that value is irrelevant to the limit computation.
- Forgetting the sign flip inside an absolute value. For $x < a$, $|x - a|$ becomes $-(x - a)$, not $(x - a)$. Missing this sign flip is the most common error on these problems.
- Arithmetic slips when plugging the breakpoint into each formula. Take an extra moment — these are usually simple substitutions, so a careless error is the main risk, not the concept itself.

4. Problem Set

Warm-up

1. $f(x) = \begin{cases} x+1, & x < 2 \\ 3x-1, & x \geq 2 \end{cases}$ — Find $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$. Does $\lim_{x \rightarrow 2} f(x)$ exist?
2. $f(x) = \begin{cases} x^2, & x < 1 \\ 2x-1, & x \geq 1 \end{cases}$ — Find both one-sided limits at $x = 1$. Does the two-sided limit exist?
3. $f(x) = \begin{cases} 3, & x < 0 \\ x+3, & x \geq 0 \end{cases}$ — Find both one-sided limits at $x = 0$. Does the two-sided limit exist?
4. $f(x) = |x|$ — Find $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$. Does $\lim_{x \rightarrow 0} f(x)$ exist?
5. $f(x) = \begin{cases} 5-x, & x \leq 3 \\ x-1, & x > 3 \end{cases}$ — Find both one-sided limits at $x = 3$. Does the two-sided limit exist?
6. $f(x) = \begin{cases} 2x, & x < -1 \\ x^2, & x \geq -1 \end{cases}$ — Find both one-sided limits at $x = -1$. Does the two-sided limit exist?

Standard

7. $f(x) = \begin{cases} 1-x^2, & x \neq 1 \\ 2, & x = 1 \end{cases}$ — Find $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1^+} f(x)$. Does $\lim_{x \rightarrow 1} f(x)$ exist, and if so, what is it?
8. $f(x) = \begin{cases} x^2-4, & x \neq 2 \\ 5, & x = 2 \end{cases}$ — Find both one-sided limits at $x = 2$. Does the two-sided limit exist?

9. $f(x) = \begin{cases} \frac{x^2-9}{x-3}, & x \neq 3 \\ 10, & x = 3 \end{cases}$ — Find $\lim_{x \rightarrow 3} f(x)$. How does it compare to $f(3)$?
10. $f(x) = \begin{cases} x+2, & x < 0 \\ x^2+2, & x \geq 0 \end{cases}$ — Find both one-sided limits at $x = 0$. Does the two-sided limit exist?
11. $f(x) = \begin{cases} 3x+2, & x \leq -2 \\ -x, & x > -2 \end{cases}$ — Find both one-sided limits at $x = -2$. Does the two-sided limit exist?
12. $f(x) = \frac{|x-3|}{x-3}$ — Find $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$. Does the two-sided limit exist?

Challenge

13. $f(x) = \begin{cases} x+3, & x < -1 \\ x^2-1, & -1 \leq x < 2 \\ 2x-1, & x \geq 2 \end{cases}$ — Find the one-sided limits at $x = -1$ and at $x = 2$, and state whether the two-sided limit exists at each point.
14. $f(x) = \begin{cases} \frac{x^2-1}{x-1}, & x < 1 \\ 3, & x = 1 \\ 2x, & x > 1 \end{cases}$ — Find both one-sided limits at $x = 1$. Does the two-sided limit exist, and how does it compare to $f(1)$?
15. $f(x) = \frac{x^2-4}{|x-2|}$ — Find $\lim_{x \rightarrow 2^-} f(x)$ and $\lim_{x \rightarrow 2^+} f(x)$. Does the two-sided limit exist?
16. Find the value of k that makes $\lim_{x \rightarrow 3} f(x)$ exist, where $f(x) = \begin{cases} kx+1, & x < 3 \\ x^2-2, & x \geq 3 \end{cases}$.
17. Find the value of c that makes $\lim_{x \rightarrow 2} f(x)$ exist, where $f(x) = \begin{cases} x^2+c, & x \leq 2 \\ 3x-4, & x > 2 \end{cases}$.

Applied

18. A (hypothetical) simplified tax model says the tax owed, $T(x)$ dollars, on income x (in thousands of dollars) is $T(x) = \begin{cases} 0.10x, & x \leq 50 \\ 0.12x-1, & x > 50 \end{cases}$. Find $\lim_{x \rightarrow 50^-} T(x)$ and $\lim_{x \rightarrow 50^+} T(x)$. Does the limit exist at the bracket boundary — in other words, is there a sudden jump in tax owed right at $x = 50$?
19. A shipping company charges $C(w) = \begin{cases} 5, & w \leq 2 \\ 5+2(w-2), & w > 2 \end{cases}$ dollars for a package weighing w pounds. Find both one-sided limits as $w \rightarrow 2$. Is there a jump in price right at the 2-pound mark?

20. A parking garage charges a flat rate per hour-block: $P(h) = \begin{cases} 5, & 0 < h \leq 1 \\ 8, & 1 < h \leq 2 \\ 11, & 2 < h \leq 3 \end{cases}$ dollars, where

h is hours parked. Find $\lim_{h \rightarrow 1^-} P(h)$ and $\lim_{h \rightarrow 1^+} P(h)$. Does the limit exist at $h = 1$? What does that tell you about the price the instant you cross into a new hour?

21. A phone data plan costs $D(x) = \begin{cases} 30, & x \leq 2 \\ 30 + 10(x - 2), & x > 2 \end{cases}$ dollars for x gigabytes used. Find both one-sided limits as $x \rightarrow 2$. Is there a sudden jump in cost right at the 2 GB mark?